

The Air and Missile War in Nagorno-Karabakh: Lessons for the Future of Strike and Defense



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The conflict between Armenia and Azerbaijan over the disputed Nagorno-Karabakh region included the heavy use of missiles, drones, and rocket artillery. The fighting, which began in late September, concluded on November 10 through a Moscow-brokered truce that resulted in the deployment of some 2,000 Russian peacekeepers and significant Armenian territorial concessions. Azerbaijan was the clear military victor, with both Russia and Turkey also benefiting politically from the war's outcome.

The 44-day war featured a diverse array of legacy and advanced air and missile strike and defense platforms. The ballistic missiles used spanned generations, from older Soviet-era Scud and Tochka missiles to the newer and more advanced Iskander and the Israeli-made LORA (Long Range Attack) missiles. Drones of Russian, Turkish, Israeli, and indigenous designs performed both reconnaissance missions to support artillery use and strike missions. Unmanned aerial vehicle (UAV) and loitering munition attacks were able to destroy heavy ground units, including T-72 tanks and

advanced S-300 air defenses. The conflict's use of these various weapons provides important information and insights into how modern wars will employ the growing spectrum of missiles, drones, and artillery.

Q1: What missiles, drones, and rockets do Armenia and Azerbaijan have?

A1: Both Armenia and Azerbaijan have invested in modernizing their militaries, including fielding more advanced air and missile systems. Azerbaijan is considered to have the more diverse and qualitatively superior military.

Armenia's missile arsenal is comprised entirely of Russian rockets. Armenia inherited its Tochka and Scud missiles from the Soviet Union following its collapse and purchased Iskander missiles from Russia in 2016. Armenia's rocket artillery is also mostly Russian, apart from its Chinese WM-80 multiple-launch rocket system (MLRS). Armenia's drone fleet consists of smaller indigenous systems focused on reconnaissance missions. They are generally recognized as less capable than Azerbaijan's fleet of foreign UAVs.

Table 1: Armenia's Missiles, Drones, and Rocket Artillery

Weapon	Notes
9K79 Tochka-U (NATO: SS-21 Scarab)	- Ballistic missile inherited from the Soviet Union. 120-kilometer (km) range. - Estimated inventory of 4 launchers.
Iskander-E (NATO: SS-26 Stone)	- Ballistic missile purchased from Russia in 2016. - Export variant of Russia's Iskander-M missile. 280-300-km range. - Estimated inventory of around 8 launchers and 25 missiles.
SS-1C Scud B	- Ballistic missile inherited from the Soviet Union. 300-km range. - Estimated inventory of around 8 launchers and 24 missiles.
X-55/Kh-55	Indigenously produced reconnaissance UAV introduced in 2014.
HRESH	Indigenously produced loitering munition introduced in 2018.
Krunk	Indigenously produced reconnaissance UAV introduced in 2011.
Orlan-10	Russian-made reconnaissance UAV. There is no record of Armenian acquisition, but reports suggest its use toward the end of the conflict.
BM-30 Smerch	300 millimeter (mm) MLRS purchased from Russia in 2015-2017. 90-km range. - Estimated inventory of 6 launchers.
Norinco WM-80	273mm MLRS purchased from China in 1999. 120 km range. - Estimated inventory of 4-8 launchers.
TOS-1A	220mm MLRS purchased from Russia in 2016. 6-10 km range.
BM-21 Grad	122mm MLRS purchased from Russia in 1995-1996. 20-km range.

Sources: Stockholm International Peace Research Institute, Ministry of Defence of Armenia.

By contrast, Azerbaijan fields a more diverse and modern arsenal of missiles, rockets, and drones. The country's oil and gas sales over the past two decades have enabled it to modernize its armed forces, including significant funding for missiles, drones, and rocket artillery. In addition to the Tochka missiles it inherited from the Soviet Union, Azerbaijan purchased the Israeli LORA ballistic missile and EXTRA (EXTended Range Artillery) guided rocket. Both are more accurate than the older Soviet missiles.

Azerbaijan also developed an impressive drone arsenal composed of Turkish and Israeli UAVs. It acquired the Turkish TB2 earlier this year, with reports suggesting the sale occurred as recently as June 2020. Previously, Azerbaijan had purchased numerous Israeli loitering munitions, also known as “suicide” or “kamikaze” drones, including the Harop, Orbiter, and SkyStriker UAVs. In the recent conflict, Azerbaijan also reportedly modified its Soviet-era An-2 Colt biplanes with remote-control systems, flying them to the front lines to draw out Armenian air defenses.

Azerbaijan likewise invested heavily in rocket artillery. The Turkish TRG-300 and Belarusian Polonez MLRS systems stand out with their ability to range targets up to 120 and 200 km away, respectively. As with Armenia, however, the BM-30 Smerch appeared to be Azerbaijan’s rocket of choice.

Table 2: Azerbaijan’s Missiles, Drones, and Rocket Artillery

Weapon	Notes
LORA	- Ballistic missile purchased from Israel in 2017-2018. 280-km range. - Accuracy of 10 meters circular error probability (CEP). - Estimated inventory of 4 launchers and 50 missiles.
9K79 Tochka-U (NATO: SS-21 Scarab)	- Ballistic missile inherited from the Soviet Union. 120-km range. - Estimated inventory of 3-4 launchers.
EXTRA	- Guided-missile purchased from Israel in 2005-2009. 150-km range with a 120-kilogram (kg) payload. - Paired with Azerbaijan's Lynx multiple rocket launcher system. - Estimated inventory of 6 launchers and 50 missiles.
Bayraktar TB2	- UAV purchased from Turkey as early as June 2020. - Commonly equipped with light munitions and can fly for 24 hours.
Harop	- Loitering munition purchased from Israel in 2014-2016. - Designed to destroy enemy radar in suppression of enemy air defenses (SEAD) missions. Also known as the Harpy 2, it can fly for up to 6 hours. - Estimated inventory of 50.
Orbiter 1K	- Loitering munition purchased from Israel in 2016-2019. - Estimated inventory of 80.
Orbiter-3	- Loitering munition purchased from Israeli in 2016-2017. - Estimated inventory of 10.
SkyStriker	- Loitering munition purchased from Israel in 2016-2019. - Two-hour flight endurance, 20-km range, equipped with a 5- or 10-kg warhead. - Estimated inventory of 100.
Hermes-900	- Medium-altitude, long-endurance (MALE) UAV purchased from Israel in 2017-2018. - Designed for intelligence, surveillance, and reconnaissance (ISR) missions. It can fly for up to 36 hours. - Estimated inventory of 2.
Hermes-450	- MALE UAV purchased from Israel in 2008-2013. - Estimated inventory of 10.
Heron	- MALE UAV purchased from Israel in 2011-2013. - Estimated inventory of 5.
Aerostar	- Surveillance UAV purchased from Israel in 2007-2012. - Estimated inventory of 14.

Searcher	• UAV purchased from Israel in 2011-2013. • Estimated inventory of 5.
Antonov An-2	• Soviet-era single-engine biplanes. • Repurposed as UAVs by equipping remote-control systems.
BM-30 Smerch	• 300mm MLRS purchased from Ukraine in 2003-2005 and Russia in 2011-2014. • 90-km range. • Estimated inventory of 30-40 launchers.
T-300 Kasirga/TRG-300 Tiger	• 300mm MLRS purchased from Turkey in 2015-2016. • 120-km range. • Estimated inventory of 20 launchers.
Belarusian Polonez	• 300mm MLRS purchased from Belarus in 2017-2019. • 200-km range. • Estimated inventory of 10 launchers.
TOS-1A	• 220mm MLRS purchased from Russia in 2011-2017. • 6-km range. • Estimated inventory of 36 launchers.
T-300	• 300mm MLRS purchased from Turkey in 2015-2016. • It can also equip TRG-300 guided rockets. • Estimated inventory of 20 launchers.
T-122	• 122mm MLRS purchased from Turkey in 2010-2014. • It can also equip TRG-300 guided rockets. • Estimated inventory of 40 launchers.
T-107	• 107mm MLRS purchased from Turkey in 2010-2013. • 11-km range. • Estimated inventory of 30 launchers.
RM-70	• 12mm MLRS purchased from the Czech Republic in 2016-2018. • Estimated inventory of 30 launchers.

Sources: Stockholm International Peace Research Institute, Ministry of Defence of Azerbaijan.

Q2: Why didn't Armenia or Azerbaijan use more longer-range missiles?

A2: Despite early concerns that fighting could escalate to the targeting of strategic infrastructure and civilian territories, both Armenia and Azerbaijan appear to have limited their use of larger, longer-range missiles. Instead, only a few events during the conflict involved ballistic missile attacks. In at least one event, Armenia reportedly

used Tochka and Scud missiles in attacks on Ganja, the second-most populous city in Azerbaijan. Azerbaijan used a LORA short-range ballistic missile in a more tactical role on October 2 to target a bridge connecting Armenia to Nagorno-Karabakh.

One potential explanation for this limited use is the small missile inventories possessed by Armenia and Azerbaijan. In contrast to the Houthi rebels in Yemen, who have relied on a steady stream of Iranian support to maintain a long ballistic missile war against Saudi Arabia, both Azerbaijan and Armenia seemed to want to conserve their limited munition stockpiles at the outset of hostilities. Both countries inherited small inventories of ballistic missiles from the Soviet Union, and each has supplemented that arsenal with more modern missiles. Armenia purchased Iskander missiles from Russia, and Azerbaijan bought the LORA from Israel. None of these sales included substantial quantities of missiles required for extended missile warfare. Despite early Armenian threats to use its more advanced Iskander missiles, the attacks on Ganja used older Soviet weapons. It was only on November 9—right before the peace agreement was signed—that footage emerged of an Armenian Iskander launch. It seems that these small arsenals forced each side to use ballistic missiles sparingly to preserve inventory if the conflict lasted longer.

A desire to contain the conflict could explain the hesitancy to use longer-range ballistic missiles. Both sides may have determined that attacks on cities or vital infrastructure may invite escalation beyond the Nagorno-Karabakh region. Furthermore, Armenia and Azerbaijan could already hit most targets in the region with long-range rocket artillery, thus limiting the value of using more expensive and limited ballistic missiles. Armenian ballistic missile strikes in Ganja, which is outside of Nagorno-Karabakh, would seem to be an outlier in this regard, though.

Azerbaijan's use of the LORA illustrates some of the limits of ballistic missiles as a tool for military operations. Baku specifically used the LORA to strike a bridge connecting Armenia to Nagorno-Karabakh in an attempt to cut off Armenian reinforcements and supplies. According to imagery after the strike, the attack failed to incapacitate the bridge, suggesting limits even to the most precise ballistic missiles. Given this apparent failure to achieve the mission with a ballistic missile, its limited arsenal, and

the alternative of cheaper rockets and drones, it is unsurprising that Azerbaijan opted to limit its use of ballistic missiles throughout the conflict.

Q3: Why did drone warfare receive so much attention?

A3: Azerbaijani drones were the center of attention in this war. Although Armenia deployed some of their own indigenously produced drones, and later footage showed their side using the more sophisticated Russian-made Orlan-10 UAV, it was Azerbaijan who took control of the skies.

As numerous recent reports have argued, these weapons were game-changing. Azerbaijani drones provided significant advantages in ISR as well as long-range strike capabilities. They enabled Azerbaijani forces to find, fix, track, and kill targets with precise strikes far beyond the front lines. UAVs were operationally integrated with fires from manned aircraft and land-based artillery but also frequently used their own ordinance to destroy various high-value military assets. Open-source reporting suggests that drones contributed to disabling a huge number of Armenian tanks, fighting vehicles, artillery units, and air defenses. Their penetration of Nagorno-Karabakh's deep rear also weakened Armenian supply lines and logistics, facilitating later Azerbaijani success in battle.

The Turkish-made Bayraktar TB2 in particular demonstrated the versatility of UAV platforms. Turkey previously used these drones to great effect in Syria and Libya. In Nagorno-Karabakh, the TB2 likewise performed well in targeting and destroying enemy defenses. In addition to providing identification and targeting data, the TB2s also carried smart, micro guided munitions to kill targets on their own. Azerbaijan has also used the high-definition cameras the TB2s carry to produce many propaganda videos. Videos showcasing attacks on Armenian fighters and equipment were posted online and broadcast on digital billboards in Baku.

Yet while drones played a large role in this conflict, their capabilities ought not be exaggerated. These platforms are very vulnerable to air defenses that are designed to counter them—defenses Armenia did not have in adequate numbers. The bulk of Armenia's air defenses consisted of obsolete Soviet-era systems, like the 2K11 Krug, 9K33 Osa, 2K12 Kub, and 9K35 Strela-10. TB2s flew too high for these systems to

intercept even if they were able to detect these relatively small aircraft. Russian-supplied Polye-21 electronic warfare systems disrupted Azerbaijani drone operations but only for four days. Armenia's Buk and Tor-M2KM air defenses likely downed a few drones, but they were deployed late in the conflict, limited in number, and vulnerable to attack themselves. Armenia's larger air defenses like the S-300 are not designed for counter-UAV missions and were targeted early in the conflict by Azerbaijani loitering munitions. According to Azerbaijan's president, Ilham Aliyev, Azerbaijani forces destroyed seven S-300 transporter erector launchers, two guidance stations, and one radar. These strikes further illustrate the vulnerability of advanced air defense systems, even if these numbers are exaggerated or the systems were not completely destroyed.

Q4: What broader lessons can we learn from the air war?

A4: The primary lesson from the air war over Nagorno-Karabakh is the importance of full-spectrum air defense. Both Armenia and Azerbaijan's short-range air defense (SHORAD) arsenals were limited in size and quality. Azerbaijan was able to exploit this gap with its large fleet of sophisticated drones.

Major powers like the United States, China, and Russia are in the process of developing and deploying their own drone countermeasures, including kinetic interceptors, electronic jammers, and even counter-drone drones. While these technologies exist today, there are difficulties in developing them at an affordable rate to provide defense at multiple echelons, including the tactical level. Armor and other heavy ground units will likely remain vulnerable until mobile SHORAD systems improve and proliferate.

The conflict also provides yet another reminder about the importance of passive defense. In an age of highly proliferated sensors and shooters, militaries will need to consider new ways to camouflage and harden their forces. Ground force tactics on dispersal and deception ought to be reinvigorated. Soldiers should train to limit their electronic and thermal signatures for longer distances and times. The video and imagery available online suggest that neither Armenian nor Azerbaijani forces had adequate resources or training on passive defense. We see this time and time again

with both sides operating out in the open, static or moving slowly; poorly camouflaged; and clumped in tight, massed formations.

The Nagorno-Karabakh conflict also illustrates that while individual weapons systems will not revolutionize the nature of warfare, the synchronization of new weapons makes the modern battlefield more lethal. Azerbaijan's combination of drones and artillery effectively targeted Armenia's high-value military assets, most notably in attacks on T-72 tanks and S-300 air defenses. In particular, strikes on air defense units constrained Armenia's ability to counter Baku's UAVs, amplifying their effectiveness. The use of UAVs and missiles to suppress and destroy air defenses gives greater validation to an observation of the U.S. Army's *Air and Missile Defense 2028* strategy: "The most stressing threat is a complex, integrated attack incorporating multiple threat capabilities in a well-coordinated and synchronized attack."

The lessons here are not new. The importance of both full-spectrum air defense and passive defenses have been shown in battles across the Middle East and in planning for potential conflict with Russia and China. The Nagorno-Karabakh conflict provides a small but important case study in the character of modern air and missile warfare.

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